

AKR1D1 (P51857) Function-Assignment Review: Does It Retain Aldose-Reductase, Ketosteroid-Monooxygenase, and Alcohol/17 β -HSD Capacities?

Gene: AKR1D1 (steroid 5 β -reductase / aldo-keto reductase family 1 member D1) **Organism:** *Homo sapiens* (NCBITaxon:9606) · **UniProt:** P51857 **Focus type:** function_assignment · **Hypothesis slug:** aldose-monooxygenase-and-alcohol-reduction-specificity

Summary

The seed hypothesis is **partially supported but over-annotated on the accessory activities**. It is **correct on its core claim and on substrate breadth**: AKR1D1's directly-supported molecular function is Δ 4-3-oxosteroid (Δ 4-3-ketosteroid) **5 β -reductase** (EC 1.3.1.3, GO:0047787), and it does reduce a broad C18–C27 range of Δ 4-3-ketosteroids. Direct homogeneous-enzyme kinetics establish this, and the seed's secondary point — that earlier negative/conflicting assays are explained by **substrate inhibition** rather than absence of activity — is well founded.

However, the seed's central proposition — that AKR1D1 "**retains**" ancestral aldose-reductase, ketosteroid-monooxygenase, and alcohol/17 β -HSD capacities *in addition to* its 5 β -reductase activity — is **not supported by direct evidence and is contradicted by structure-mechanism reasoning**. These accessory GO terms rest on weak provenance: phylogenetic inference (IBA:GO_Central), a single reaction-record electronic annotation (IEA:RHEA), a reaction-class mislabel from Reactome (TAS), or a paralog expression study that does not even assay AKR1D1. Critically, AKR1D1 carries a **His→Glu120 substitution** at the position occupied by the catalytic histidine of every general carbonyl-reducing AKR (including aldose reductase). Glu120 is the residue AKR1D1 *requires* for its unique β -face hydride transfer to the Δ 4 C=C double bond, and it mechanistically **disfavors** the carbonyl $\leftrightarrow$ alcohol chemistry that aldose

reductase, alcohol dehydrogenase, and 17 β -HSD demand. Ketosteroid monooxygenase is chemically impossible for an NADPH oxidoreductase, since it requires O₂-dependent Baeyer–Villiger oxygenation.

The appropriate curation posture is therefore to **retain the core 5 β -reductase MF** and to **flag the accessory terms for removal, NOT-qualification, or downgrade to non-core**, pending curator verification. The seed's methodological warning — "do not infer loss of ancestral capacity from predominant steroid use or target-assay absence alone" — is valid, but here the evidence is not merely *absence of assay*; it is a **positive mechanistic reason** (the catalytic-His replacement) plus **traceable database carry-over** for each accessory term.

Executive Judgment

Verdict: Partially supported — over-annotated on aldose-reductase, ketosteroid-monooxygenase, alcohol-dehydrogenase, and 17 β -HSD; core 5 β -reductase strongly supported.

The single most important structural fact is that AKR1D1 has repurposed the AKR catalytic histidine into Glu120, which is essential for its double-bond reduction chemistry but incompatible with efficient aldehyde/ketone carbonyl reduction. Every accessory annotation either requires that lost carbonyl chemistry (aldose reductase, alcohol dehydrogenase, 17 β -HSD) or requires oxygenase chemistry an AKR cannot perform (ketosteroid monooxygenase). The provenance for each accessory term is weak, and the AKR1C IDA source the seed relies on assays the paralogs, not AKR1D1.

Most important caveat: No study has directly assayed purified AKR1D1 against canonical aldose or monooxygenase substrates and formally reported a null. The exclusion is inferred from structure, sequence, and GO/reaction definitions — a strong but not airtight basis. The one accessory term that most warrants a full-text re-read before a hard NOT is 17 β -HSD (RHEA:53484, from PMID11342103), because its intact-cell background could mask whether the observed 17 β -hydroxy product reflects intrinsic AKR1D1 activity.

Key Findings

F001 — The His→Glu120 substitution mechanistically disfavors general carbonyl reduction

The aldo-keto reductase (AKR) superfamily performs carbonyl reduction using a conserved catalytic tetrad of **Asp–Tyr–Lys–His**. In AKR1D1, UniProt P51857 confirms that Asp53, Tyr58, and Lys87 are present, but the conserved catalytic **histidine is replaced by glutamate at position 120** (residue 120 = E in AKR1D1, whereas the aligned His117 = H in the paralog AKR1C2). This single substitution is not incidental: it is the molecular basis of AKR1D1's distinct chemistry.

Di Costanzo, Drury, Penning & Christianson (2008; [PMID: 18407998](#)) solved the crystal structure and demonstrated that the catalytic dyad is **Tyr58 + Glu120**, and that both **Y58F and E120A mutants are devoid of activity**. Glu120 enables β -face hydride transfer onto the Δ^4 C=C double bond via an enolate intermediate — a reaction chemistry that is *unique* among AKRs and fundamentally different from reducing a free aldehyde or ketone carbonyl. The verified quote:

"Each steroid carbonyl accepts hydrogen bonds from catalytic residues Tyr(58) and Glu(120). The Y58F and E120A mutants are devoid of activity, supporting a role for this dyad in the catalytic mechanism."

This is the linchpin structural argument: the residue the seed hypothesis would need to invoke for general carbonyl chemistry (the catalytic His) has been repurposed into the residue (Glu120) that makes AKR1D1 a double-bond reductase. The seed correctly notes that PMID18407998 is "not a blanket exclusion of additional chemistry" — a single active-site study cannot exhaustively rule out trace promiscuous side reactions. But it does establish a *positive* mechanistic reason to doubt efficient aldose/ketone carbonyl reduction, which is stronger than mere silence.

F002 — Core 5 β -reductase function is robustly supported across broad substrates; multifunctional labels rest on weak/indirect evidence

Chen, Drury & Penning (2011; [PMID: 21255593](#)) used **homogeneous (purified) enzyme** to show that AKR1D1 reduces all C18, C19, C21, and C27 Δ^4 -3-ketosteroids tested at physiological pH. Crucially, they observed **substrate inhibition** with C18–C21 steroids when the C11 position is unsubstituted, explaining why earlier assays (e.g., Kondo et al. 1994, [PMID: 7508385](#)) sometimes reported weak or negative activity for particular substrates. The verified quote:

"AKR1D1 proficiently reduced all the steroids tested at physiological pH, indicating AKR1D1 is the only enzyme necessary for all the 5 β -steroid metabolites present in humans. Substrate inhibition was observed with C18 to C21 steroids provided that the C11 position was unsubstituted."

The physiological centrality of this activity is independently confirmed by human disease genetics: biallelic loss-of-function variants cause **Congenital Bile Acid Synthesis defect type 2 (CBAS2)**, presenting with neonatal cholestasis, coagulopathy, and failure to thrive ([PMID: 41387259](#); [PMID: 26418565](#), the P133R mutation). These confirm that the Δ^4 -3-ketosteroid 5 β -reduction step in bile-acid synthesis is AKR1D1's essential in-vivo role.

By contrast, each accessory annotation has thin provenance: **17 β -HSD (GO:0072582)** is only IEA:RHEA from a single reaction record (RHEA:53484); **alcohol dehydrogenase, NADP+ (GO:0008106)** is only TAS:Reactome; **aldose reductase (GO:0004032)** and **ketosteroid monooxygenase (GO:0047086)** are only IBA:GO_Central (phylogenetic inference).

The seed asserts that the AKR1C IDA multifunctionality traces to [PMID: 21232532](#). That paper is an **endometriosis expression study** that assays AKR1C1/AKR1C2/AKR1C3 (the paralogs) — not AKR1D1 — and reports **negligible AKR1D1 mRNA**:

"significantly increased mRNA levels of AKR1C1, AKR1C2, AKR1C3 and SRD5A1, and negligible mRNA levels of AKR1D1"

An expression study of the paralogs that finds AKR1D1 barely expressed cannot transfer 17 β /20 α -HSD multifunctionality onto AKR1D1. This is the clearest example of **paralog carry-over** in the annotation set.

F003 — The Reactome "alcohol dehydrogenase" label maps entirely to Δ^4 -double-bond reductions, not C-OH oxidation

Querying the Reactome ContentService for UniProt P51857 returns exactly **five reactions** for AKR1D1, all within the three bile-acid synthesis pathways (R-HSA-193368 / 193775 / 193807). Every one of the five (R-HSA-192033, 192067, 193746, 193821, 193824) has the form "**4-cholesten-...-3-one is reduced to 5 β -cholestan-...-3-one**" — i.e., the stereospecific reduction of the Δ^4 C=C double bond of a 3-oxo bile-acid intermediate to the 5 β product. **None is an alcohol (C-OH) oxidation or reduction.**

Therefore the TAS:Reactome GO:0008106 ("alcohol dehydrogenase, NADP+") annotation is a **reaction-class mislabel** applied to the 5 β -reductase reaction — not evidence of a distinct alcohol dehydrogenase activity. The seed hypothesis explicitly asked to "distinguish this from Reactome alcohol-dehydrogenase labels for actual Δ^4 -double-bond reductions," and the Reactome content resolves the question directly in favor of *mislabel*, not *genuine additional activity*.

F004 — Pairwise alignment confirms AKR1D1 Glu120 occupies the exact position of aldose reductase's catalytic His

A Needleman–Wunsch global alignment of AKR1D1 (P51857) against human aldose reductase AKR1B1 (P15121) gives **52.5% identity** — consistent with the ~50% cited historically (PMID7508385). The catalytic residues co-align cleanly: AKR1D1 Tyr58 $\leftrightarrow$ AKR1B1 Tyr49 (the proton donor), Asp53 $\leftrightarrow$ Asp44, and Lys87 $\leftrightarrow$ Lys78 — all conserved. But at the decisive position, **AKR1D1 Glu120 aligns to AKR1B1 His111**, the catalytic histidine of aldose reductase. The local motif contrast is diagnostic: AKR1B1 "YLIHWPT" vs AKR1D1 "YIIEVPM" — the His $\rightarrow$ Glu substitution *and* loss of the adjacent Trp (Trp $\rightarrow$ Val).

This is a direct sequence-level confirmation of the structural argument in F001: the residue aldose reductase uses for aldehyde/aldose carbonyl reduction is precisely the residue AKR1D1 has substituted. It makes the ancestral **aldose reductase (GO:0004032)** activity mechanistically unlikely to be retained at a physiologically meaningful level.

F005 — GO term definitions confirm the accessory annotations require chemistry AKR1D1 cannot perform

Reading the QuickGO definitions of the accessory terms against AKR1D1's mechanism shows a systematic mismatch:

GO term	Definition (reaction)	Chemistry required	AKR1D1 capability
GO:0047086 ketosteroid monooxygenase	$O_2 + \text{NADPH} + \text{progesterone} = \text{H}_2\text{O} + \text{NADP}^+ + \text{testosterone acetate}$	O₂-consuming Baeyer–Villiger monooxygenation	None — AKR1D1 is an NADPH oxidoreductase, not an oxygenase; chemically impossible
GO:0004032 aldose reductase	$\text{alditol} + \text{NAD(P)}^+ = \text{aldose} + \text{NAD(P)H}$	Carbonyl↔alcohol using catalytic His	His replaced by Glu120 (F001/F004)
GO:0008106 alcohol dehydrogenase (NADP+)	$\text{alcohol} + \text{NADP}^+ = \text{aldehyde/ketone} + \text{NADPH}$	Carbonyl↔alcohol using catalytic His	His replaced by Glu120; Reactome reactions all $\Delta 4$ reductions (F003)
GO:0072582 17 β - HSD	$17\beta\text{-hydroxysteroid} + \text{NADP}^+ = 17\text{-oxosteroid} + \text{NADPH}$	C17 carbonyl↔alcohol redox	His replaced by Glu120; single IEA:RHEA record
GO:0047787 $\Delta 4$-3- oxosteroid 5β- reductase (core)	reduces progesterone, androstenedione, 17 α -OH-progesterone, testosterone, bile-acid intermediates	β-face hydride transfer to $\Delta 4$ C=C	Yes — established (F002)

The **ketosteroid monooxygenase** term is the most decisively excludable: no aldo-keto reductase can perform an O₂-dependent Baeyer–Villiger oxygenation converting progesterone to testosterone acetate. The remaining three accessory terms all require the carbonyl↔alcohol interconversion chemistry that depends on the catalytic His that AKR1D1 has repurposed as Glu120.

Mechanistic Model / Interpretation

The unifying mechanistic story is a **single active-site substitution that redirects catalysis**:

AKR superfamily (general carbonyl reductases, e.g. AKR1B1, AKR1C):

catalytic tetrad = Asp - Tyr - Lys - HIS

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+--> protonates/positions substrate CARBONYL
-> aldose reductase, 17b-HSD, alcohol DH chemistry

AKR1D1 (steroid 5b-reductase):

catalytic tetrad = Asp53 - Tyr58 - Lys87 - GLU120 (HIS -> GLU)

|
+--> enables b-face HYDRIDE transfer to the
Delta4 C=C double bond (enolate mechanism)
-> Delta4-3-ketosteroid 5b-reduction ONLY

Everything downstream follows from this. AKR1D1 sits near the top of the aldo-keto reductase family tree that also produced the general carbonyl reducers, so **phylogenetic (IBA) pipelines and paralog-based annotation transfer** naturally propose that AKR1D1 "retains" ancestral aldose-reductase and monooxygenase activities. But AKR1D1 is a *specialist* whose defining innovation (Glu120) is chemically incompatible with efficient carbonyl reduction. The five Reactome reactions confirm that in the human bile-acid pathway, AKR1D1 does exactly one kind of chemistry — $\Delta^4 \rightarrow 5\beta$ reduction of 3-oxo bile-acid intermediates — regardless of whether a given database labels that step "alcohol dehydrogenase" or "5 β -reductase."

The seed hypothesis makes a legitimate epistemic point ("do not infer loss from predominant steroid use alone"), and it is correct that absence-of-assay is not proof-of-absence. The resolution is that this case does **not** rest on absence of assay: it rests on (1) a positive structural/mechanistic reason (His $\rightarrow$ Glu120), (2) sequence-level confirmation that Glu120 sits at aldose reductase's catalytic-His position, (3) GO definitions that demand incompatible chemistry, and (4) traceable database carry-over for every accessory term. That is a stronger basis for down-weighting the accessory activities than "we didn't look."

Bottom line: AKR1D1 = Δ^4 -3-oxosteroid 5 β -reductase (specialist). The accessory aldose/monooxygenase/ADH/17 β -HSD labels are over-annotations arising from phylogenetic inference, reaction-class mislabels, and paralog confusion — not from demonstrated AKR1D1 activities.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding
PMID: 18407998	Structural (crystal + mutagenesis)	Refutes accessory carbonyl activities; supports core	What is the catalytic dyad; is the catalytic His present?	Catalytic dyad Glu120; Y581 devoid of ac enables β -fa transfer to Δ
PMID: 21255593	Direct assay (homogeneous enzyme kinetics)	Supports core; qualifies older negatives	Substrate scope of 5β -reduction; cause of prior negatives	Reduces all C ketosteroids pH; substrat unsubstituted explains earl
PMID: 7508385	Direct assay (historical)	Qualifies	Broad $\Delta 4$ -3- ketosteroid range; some negatives	Broad steroid range; some attributed to inhibition
PMID: 11342103	Direct assay (intact-cell context)	Qualifies/ competing	Basis of RHEA:53484 17- keto $\rightarrow$ 17 β -OH	Underlies Un 17 β -HSD and product/posi cell backgro scrutiny
PMID: 21232532	Expression (mRNA)	Refutes paralog transfer	Does the cited AKR1C IDA source apply to AKR1D1?	Assays AKR1 AKR1D1; rep AKR1D1 mR
PMID: 26418565	Mutant phenotype + kinetics	Supports core physiological role	Is 5β -reduction the essential in- vivo function?	P133R impair binding/hydr bile acid def
PMID: 41387259				

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding
	Mutant phenotype (clinical)	Supports core physiological role	Physiological consequence of AKR1D1 loss	Biallelic LOF absent prim fatal infant c
Reactome (R-HSA-192033/192067/193746/193821/193824)	Database (pathway)	Refutes ADH label as distinct activity	Do "alcohol dehydrogenase" labels denote C-OH chemistry?	All 5 AKR1D1 $\Delta 4 \rightarrow 5\beta$ reduc bile-acid inte is C-OH oxid
QuickGO term definitions	Database (ontology)	Refutes accessory MF terms	Do accessory GO defs match AKR1D1 chemistry?	GO:0047086 Baeyer-Villig GO:0004032/ need catalyt redox
UniProt P51857 + AKR1B1 (P15121) alignment	Computational (sequence)	Refutes aldose reductase retention	Does Glu120 occupy the aldose-reductase catalytic-His site?	52.5% identi Glu120 $\leftrightarrow$ His conserved; a

GO Curation Implications

Core MF term — retain. - **GO:0047787** ($\Delta 4$ -3-oxosteroid 5β -reductase activity, MF) is strongly supported by direct homogeneous-enzyme kinetics (PMID21255593), structure/mechanism (PMID18407998), and disease genetics (PMID26418565, PMID41387259). **Retain as the primary/core molecular function.** - Associated BP: bile acid biosynthetic process (and steroid metabolic process) is well supported by CBAS2 disease genetics — **retain**.

Accessory MF terms — flag for curator action (leads requiring verification):

GO term	Current evidence	Recommended lead action
GO:0047086 ketosteroid monooxygenase	IBA:GO_Central	Remove / NOT — requires O ₂ -dependent Baeyer–Villiger oxygenation impossible for an AKR (F005). Highest-confidence removal.
GO:0004032 aldose reductase	IBA:GO_Central	Remove or downgrade to non-core — catalytic His replaced by Glu120 (F001/F004); ancestral inference only, no AKR1D1 direct assay.
GO:0008106 alcohol dehydrogenase (NADP+)	TAS:Reactome	Remove / re-map — Reactome reactions are all $\Delta^4 \rightarrow 5\beta$ reductions, a reaction-class mislabel, not C-OH oxidation (F003).
GO:0072582 17 β -HSD	IEA:RHEA (RHEA:53484)	Downgrade to non-core / verify — single reaction record from PMID11342103 in an intact-cell background; product/position may reflect a different steroid transformation. Treat as uncertain pending re-read.

Avoid "protein binding" as a fallback — a specific, informative MF (GO:0047787) is well supported, so no generic term is needed.

Mechanistic Scope

The immediate molecular function under test is **NADPH-dependent reduction of the Δ^4 C=C double bond of 3-oxosteroids to yield 5 β -dihydro products** (β -face hydride transfer via an enolate intermediate, gated by Tyr58/Glu120). This is a *direct catalytic activity* of the AKR1D1 gene product.

Distinguished from this direct activity: - **Downstream pathway consequence:** production of 5 β bile-acid intermediates and, ultimately, primary bile acids (cholic/chenodeoxycholic acid). A pathway output, not a separate molecular function. - **Disease manifestation (loss of function):** CBAS2 — neonatal cholestasis, coagulopathy, failure to thrive, potential fatal outcome. Phenotypes of enzyme deficiency, not evidence of additional enzymatic activities. - **Inferred/ancestral activities (aldose reductase, ketosteroid monooxygenase):** *evolutionary*

inferences transferred from ancestral or paralogous AKRs, not measured AKR1D1 activities. - **Reaction-class labels (alcohol dehydrogenase):** database annotations attached to the same 5 β -reductase reactions, not distinct chemistries.

Conflicts and Alternatives

1. **Paralog confusion (strongest confound).** The seed traces AKR1C1/C2/C3 IDA multifunctionality to PMID21232532, but that paper assays the AKR1C paralogs and reports *negligible AKR1D1 expression*. Any transfer of 17 β /20 α -HSD activity from AKR1C to AKR1D1 is paralog carry-over, not direct evidence.
 2. **Reaction-class mislabel.** The Reactome "alcohol dehydrogenase (NADP+)" descriptor is applied to reactions that are chemically $\Delta^4 \rightarrow 5\beta$ reductions. A curator reading only the GO label could mistake this for a genuine second activity.
 3. **Database carry-over via phylogenetic inference.** GO:0004032 and GO:0047086 are IBA (inferred from biological ancestor). AKR1D1's ancestral position in the AKR tree makes such inference *expected* but not *correct* given the His $\rightarrow$ Glu120 innovation.
 4. **Intact-cell / product-identity ambiguity (17 β -HSD).** RHEA:53484 (from PMID11342103) records a 17-keto $\rightarrow$ 17 β -hydroxy conversion. In an intact-cell background, the observed product could arise from a coupled activity or paralog rather than intrinsic AKR1D1 17 β -HSD chemistry. This one warrants a full-text re-read before any hard NOT call.
 5. **The seed's valid counter-argument.** "Do not infer loss from predominant steroid use or target-assay absence alone" is methodologically sound. The rebuttal is that the case rests on *positive* mechanistic and provenance evidence, not on assay silence — but a curator should note that no study has *directly assayed and formally excluded* trace aldose-reductase or monooxygenase activity of purified AKR1D1.
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Limitations and Knowledge Gaps

1. **No direct negative assay for aldose/monooxygenase activity on purified AKR1D1.** PMID21255593 assayed steroids, not aldoses or a Baeyer–Villiger reaction. The exclusion is inferred from structure/mechanism and definitions, not from a targeted null. *Resolution:* assay purified AKR1D1 against canonical aldose-reductase substrates (glyceraldehyde,

glucose) and against the GO:0047086 progesterone→testosterone-acetate reaction with O₂/NADPH.

2. **PMID11342103 (17β-HSD / RHEA:53484) not read in full.** The product position (C17) and the intact-cell background remain unverified. This is the single accessory term where a genuine (if minor) activity cannot yet be firmly excluded. *Resolution:* obtain and read the full text; determine whether the 17β-OH product reflects intrinsic AKR1D1 activity or a coupled/paralog contribution.
3. **PMID21232532 read from abstract snippet only.** "Negligible AKR1D1 mRNA" is from the abstract; the enzymology attributed to AKR1C is not AKR1D1's regardless, so this gap is low-risk for the conclusion.
4. **Alignment is global pairwise, not a full structural superposition.** The Glu120↔His111 positional homology is high-confidence but sequence-derived; a structural superposition of AKR1D1 (e.g., PDB 3BUR/3COT) onto AKR1B1 would make it definitive.
5. **Substrate-inhibition kinetics not independently reproduced here.** The claim that substrate inhibition resolves older negatives rests on PMID21255593; not independently re-derived.

Discriminating Tests

The following would most efficiently separate "retained ancestral activities" from "specialist 5β-reductase over-annotated":

1. **Targeted enzymology on purified recombinant AKR1D1** against:
2. **Aldose reductase substrates** (DL-glyceraldehyde, glucose) + NADPH → measures GO:0004032 directly. Predicted: negligible/absent.
3. **Ketosteroid monooxygenase reaction** (progesterone + O₂ + NADPH → testosterone acetate) → measures GO:0047086 directly. Predicted: absent (no oxygenase chemistry).
4. **17β-HSD reaction** (17-keto ↔ 17β-hydroxy steroid), cell-free → tests whether RHEA:53484 reflects intrinsic activity. Predicted: absent or negligible relative to 5β-reduction.
5. **E120→H "back-mutation" gain-of-function test.** Introduce Glu120His and assay for gain of general carbonyl-reduction activity. If the His→Glu substitution is the specificity switch, restoring His should confer measurable carbonyl reduction while abolishing 5β-reduction.
6. **Structural superposition** of AKR1D1 (PDB 3BUR) onto AKR1B1 (PDB 1US0) to confirm Glu120 occupies the aldose-reductase catalytic-His pocket in 3D.

7. **Full-text extraction of PMID11342103** to determine the exact product, position, and whether the assay was cell-free — the decisive check for the 17β -HSD term.
8. **Provenance trace of RHEA:53484 → UniProt IEA** to confirm the annotation is a single-reaction electronic inference rather than curated experimental support.

Curation Leads (require curator verification)

Retain (core): - **GO:0047787** Δ^4 -3-oxosteroid 5β -reductase activity (MF) — supported by [PMID: 21255593](#), [PMID: 18407998](#), [PMID: 26418565](#). - *Snippet to verify (PMID21255593):* "AKR1D1 proficiently reduced all the steroids tested at physiological pH... Substrate inhibition was observed with C18 to C21 steroids provided that the C11 position was unsubstituted." - BP: bile acid biosynthetic process; steroid metabolic process — supported by CBAS2 genetics ([PMID: 41387259](#)).

Flag for removal / NOT (leads): - **GO:0047086** ketosteroid monooxygenase (IBA) → *remove/NOT*. Requires O_2 -dependent Baeyer–Villiger oxygenation; AKR1D1 is an NADPH oxidoreductase. - **GO:0004032** aldose reductase (IBA) → *remove or non-core*. Catalytic His replaced by Glu120 (F001/F004); ancestral inference only. - **GO:0008106** alcohol dehydrogenase, $NADP^+$ (TAS:Reactome) → *remove/re-map*. All AKR1D1 Reactome reactions are $\Delta^4 \rightarrow 5\beta$ reductions, not C-OH oxidation (F003).

Downgrade / verify (lead): - **GO:0072582** 17β -HSD (IEA:RHEA, RHEA:53484) → *non-core / verify*. Requires full read of [PMID: 11342103](#); product position and intact-cell background must be checked before a hard NOT.

Suggested curator questions: 1. Has purified AKR1D1 ever been directly assayed and found negative for aldose-reductase or monooxygenase activity, or is the exclusion purely mechanistic/inferential? 2. Does RHEA:53484 (17β -HSD) derive from an intact-cell or cell-free assay in PMID11342103? 3. Should IBA terms conflicting with an experimentally defined specialist function be auto-flagged for review?

Suggested experiments: the E120H back-mutation gain-of-function assay and the targeted aldose/monooxygenase enzymology in "Discriminating Tests."

Evidence Base (literature synthesis)

- *Crystal structure of human liver Δ^4 -3-ketosteroid 5 β -reductase (AKR1D1) and implications for substrate binding and catalysis.* [PMID: 18407998](#) — Establishes Tyr58/Glu120 catalytic dyad; Y58F and E120A null. **Core structural pillar** for the specificity argument.
 - *Substrate specificity and inhibitor analyses of human steroid 5 β -reductase (AKR1D1).* [PMID: 21255593](#) — Homogeneous-enzyme kinetics across C18–C27 Δ^4 -ketosteroids; substrate inhibition resolves older negatives. **Core functional pillar.**
 - *In-Depth Dissection of the P133R Mutation in Steroid 5 β -Reductase (AKR1D1).* [PMID: 26418565](#) — Disease mutation impairs NADPH binding/hydride transfer → bile acid deficiency. Confirms core physiological role.
 - *Variants in AKR1D1 and Infant Mortality.* [PMID: 41387259](#) — Biallelic LOF causes CBAS2 with near-absent primary bile acids. Confirms in-vivo essentiality of the 5 β -reduction step.
 - *AKR1D1 and CYP7B1 mutations in inborn errors of bile acid metabolism.* [PMID: 31337596](#); infant cholestasis case reports [PMID: 30254413](#), [PMID: 36739965](#), [PMID: 38034430](#) — Consistent disease genetics reinforcing the bile-acid-synthesis role.
 - *Endometriosis expression study.* [PMID: 21232532](#) — Assays AKR1C1/2/3, reports negligible AKR1D1 — refutes paralog transfer of multifunctionality.
 - *17 β -conversion source.* [PMID: 11342103](#) — Basis of RHEA:53484 17 β -HSD annotation; requires full-text scrutiny (open gap).
 - Database/computational: Reactome ContentService (5 Δ^4 →5 β reactions), QuickGO term definitions, UniProt P51857, and AKR1D1↔AKR1B1 pairwise alignment (52.5% identity; Glu120↔His111).
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Conclusion

The seed hypothesis is **right about the core and about substrate breadth**, and it makes a fair methodological point about not inferring loss from assay absence. But its central claim — that AKR1D1 *retains* aldose-reductase, ketosteroid-monooxygenase, and alcohol/17 β -HSD capacities as genuine additional functions — is **not supported by direct evidence and is contradicted by a coherent structure–mechanism–provenance chain**. AKR1D1 is a specialist Δ^4 -3-oxosteroid 5 β -reductase whose defining Glu120 substitution repurposes the AKR catalytic histidine; the Reactome "alcohol dehydrogenase" label is a reaction-class mislabel for the same

5 β reduction; ketosteroid monooxygenase is chemically impossible for an oxidoreductase; and the AKR1C IDA source assays the paralogs, not AKR1D1. The accessory GO terms should be flagged for removal, NOT-qualification, or downgrade to non-core — with the 17 β -HSD (RHEA) term the one that most warrants a full-text re-read before a hard call.

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